Etymology and the European Lexicon

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Latin *gingīva* and *salīva*

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The traditional etymology of Lat. *gingīva* 'gum(s)' (Afran.+) via the questionable PIE root **geng-* 'lump(y), ball' (Solmsen 1909: 219, IEW, WH) is to be rejected; the word thus has no satisfactory etymology (so de Vaan, EDLIL s.v.). Comparison with words for 'palate; gum' in Germanic and Baltic (e.g., OE *gōma* 'palate' > Eng. *gum*), with a well-established background in PIE **ĝʰeh₂-(u-)* 'gape', suggests an interpretation of *gingīva* on the basis of a zero grade belonging to the parallel "*i*-present" **ĝʰeh₂-i-* 'id.' (cf. Lat. *hiāre, hīscere* 'id.'). Further comparison with a related word for 'mouth' (PIE **ĝʰh₂ói̯-u-* > TB *koyn,* TA *koy-*; Pr.-Sl. **zěvŭ*) indicates that *gingīva* may be a compound or univerbation involving a *figura etymologica* meaning (roughly) 'open-mouthed [stuff]' (perhaps originally referring to the palate, as in Gmc.), comparable to the formulaic *figura etymologica* TB *koyn kāy-* 'open the mouth'. This analysis invites reconsideration of the background of Lat. *salīva* 'spittle; slimy substance' (Varro+), whence a proposal to revive (and modernize) the etymology (e.g., Vaniček 1881: 341f.) based on the root of Lat. *līmus* 'mud, slime', OIc./OE *slīm* 'slime', OCS *sliny* 'saliva'.

1 Although first attested in Afranius,[1] and then, somewhat later, in Catullus (39.19, 97.6), Lat. *gingīva* 'gum(s)' (mainly used in the plural) may be old within Latin. As an anatomical term ill-suited for most poetic genres[2] and other forms of high discourse, the word is mainly found in technical writings on medicine.[3] Thus its non-appearance in earlier texts – even the "health care" portions of Cato *Ag.* or the body-part-rich abuse of Plautus – could well be a matter of chance. Apart from a diminutive *gingīvula*,[4] the word has no derivatives in the Classical language, and its only comparand is the similar (or at least similar-looking) suffixation in *salīva* 'saliva, spittle; slimy or viscous substance' (Varro+), as often noted.[5]

2 The native etymological tradition offers explanations of the word based on *gignō* 'bear, (pass.) be born' and (with spelling *gengiva*) *genū* 'knee'; but these have no interest except as folk etymologies.[6]

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1 *togat.* 58 Ribb. (*ex coni.*, but the emendation is virtually certain; see Ribbeck ad loc. and Daviault 1981: 158 ad loc.); thus correct the etymological dictionaries (WH "seit Catull.", similarly EM and EDLIL, s.v.).

2 Apart from abuse: see, for example, Katz 2000 on Catullus 39. Besides Catullus and Afranius, the only other literary usages (all in unflattering contexts) appear in Juvenal (10.20), Apuleius (*Met.* 8.23), and Fronto (*Am.* 1.12.2 [178.19 van den Hout] = Haines 2.172).

3 Frequently in Celsus and Pliny the Elder; attestations also in Dioscorides, Scribonius Largus, Marcellus, Palladius, Quintus Serenus, Soranus, and other medical writers, as well as in glosses (see TLL s.v.).

4 Attested with certainty only in Apuleius (*Apol.* 6), perhaps as a coinage.

5 There is no significant MS variation: the spelling *geng-* (Pelagonius, Soranus, etc.) may be partly influenced by a folk etymology (see next note), and the medieval variants of the type *gingib-, gingif-, gyngiv-*, etc., along with still more extreme deformations in glossators, preserve nothing of value. (See TLL s.v. for details.)

6 "a gignendis dentibus nominatur" (Lactantius, Isidore); "quod iuxta genua in utero formentur" (Vindicianus). Again, see TLL for details.

The closest thing to a standard modern etymology is that of Solmsen (1909: 219),[7] who compared a series of words meaning 'lump; round; ball' and the like,[8] suggesting that *gingīva* is elliptical for an original adjective phrase *gingīva* [scil. *carō* 'flesh'],[9] with *gingīva* based on a noun **gengā* 'Beule, Buckel' or an adjective **gengos* 'kompakt, massig, fest'. There is in any case no PIE word for this body part, and no evidence that "PIE **geng-*" (if such a PIE root can really be reconstructed) was used in body-part terms; so one is reduced to typological comparisons for the semantic analysis. These are not entirely lacking, as Schwyzer has discussed (1930). Thus OCS *dęsna* [scil. *plŭtĭ* 'flesh'] 'gum' supports the possibility of an elliptical phrase with *carō*; yet *dęsna* itself is almost certainly based on the word for 'tooth',[10] and thus offers no support for Solmsen's root and semantic analysis of *gingīva.* For this, Schwyzer (1930: 265ff.) adduced OHG *bilarn,* NHG dial. *Billern* 'gum', related to forms like NHG *Ball* 'ball' and *Bolle* 'onion' – certainly suggestive, but hardly a compelling confirmation of Solmsen's analysis.

Solmsen's **gengā* and **gengos* also raise formal concerns: these are purely theoretical constructs, as *e*-grades to this root are virtually unattested.[11] Lat. *gingīva* could in theory reflect a zero grade **gn̥g$^{(h)}$-,* to which one might compare some Baltic material with *u*-vocalism (Li. *gùnga* 'hunch, lump', etc.);[12] but as de Vaan points out (EDLIL s.v.), these forms might have old **u* (see LEW s.v. *gugà*). Nor do Solmsen's preforms provide any accounting for the suffixation in *-īva* except by analogy to *salīva,* the formation of which is itself poorly understood (a point to which we will return in some detail). Moreover, no evidence (apart from presumptions about the morphological background of the suffix *-īva*) suggests that Lat. *gingīva* was an original adjective in an elliptical expression with *carō*, and indeed the plural usage, securely attested from the time of Catullus, argues against this.[13] It is also worth recalling that the meaning 'flesh', for Lat. *carō,* is a late development, since the corresponding Sabellic terms (O. *carneis,* U. **karu** 'section, part') lack this specialization.[14] Thus the ellipsis in question would imply that *gingīva* is a relatively late formation – highly unlikely for such an opaque and isolated body-part term, for which one might rather expect a better Indo-European pedigree. Instead, we have a possibly fanciful metaphor involving 'lumping' or 'compaction', based on a "root" that itself is often considered to be fundamentally expressive and with a dubious PIE status in the first place.[15]

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7 Accepted, e.g., by IEW (see next note), GEW (s.v. *γογγύλος*), and apparently also by WH I.601 s.v. (but see further below); still cited (though tentatively) by Meiser (1998: 83).

8 IEW 379f.: **geng-*, **gong-* 'Klumpen, klumpig', **gongu-s* 'Ball' (nearly all with *o*-grade: Gk. *γογγύλος* 'round', etc.), possibly including late Germanic forms that refer to 'bending' and 'winding' (MLG *kinke,* Norw. dial. *kink*(*e*) 'Windung bei einem Tau', etc.), and with a root variant **gonĝ-* in Balto-Slavic (IEW 380).

9 Cf. (according to Solmsen) *salīva* [scil. *aqua*].

10 See recently EDSIL 107 (s.v. **dḗsnò/à*); compare similar compound expressions for 'gum(s)' of the type NHG *Zahnfleisch,* Skt. *danta-māṃsa-* (Suśruta), both literally 'tooth-flesh'.

11 Only Gk. *γιγγίς, γιγγίδιον* 'type of beet' (allegedly assimilated from **γεγγι-*); but Beekes (EDG 281, s.v. *γογγύλος*) considers the relationship of this word to *γογγύλος* to be improbable. The *e*-grade forms belonging to the late Gmc. 'winding' forms (n. 8 above) have little value: de Vaan (EDLIL 262) convincingly rejects, on semantic grounds, any connection with this material, a point that may be extended to **geng^{h}-* 'drehen, winden, flechten, weben' (IEW 380, Germanic and Slavic only).

12 Thus (tentatively) Meiser, loc. cit. (n. 7 above).

13 Nor, for that matter, does Solmsen's "*salīva* [scil. *aqua*]" rest on any evidence at all.

14 See on this point Schwyzer 1930: 263f.

15 On the expressive character of the **gong-*/**g*(*e*)*ng-* material, see GEW loc. cit. (n. 7 above) and DELG (s.v. *γογγύλος*). Note also Beekes (EDG loc. cit., n. 11), who judges the etymology of Gk. *γογγύλος* to be "unknown" and its connection with Gmc. **gongu-* and Li. *gungulỹs* 'ball' to be "highly doubtful".

Despite the lip-service paid by WH to Solmsen's theory, one detects a certain lack of confidence in it, to judge from their "Nicht besser …", introducing the alternative attempt (Ribezzo, Schwyzer)[16] to etymologize the word via Lat. *cingō* 'gird', i.e. (originally) **cingīva [carō]* 'Fleisch, das (die Zähne) umgürtet". Again, one can bolster the semantic plausibility of such a theory on the basis of typological comparanda.[17] But WH offer a convincing refutation, observing first that "*cingere* 'umgürten, umschließen' niemals in Wendungen wie *carō cingit dentēs* gebraucht wird", and secondly that the alleged support from Romance **gincīva* (Fr. *gencive,* etc.) is spurious, since this form arises from a dissimilated *gingīva* and not a metathesized **cingīva.*[18] The entry goes on, finally, to list other proposals that are even more desperate ("Sicher abzulehnen …") and which need not detain us here.[19]

As often, Meillet (in EM 275, s.v.) provides the most sensible synoptic judgment: "Il n'a été fait que des rapprochements vagues", adding that "[l]a forme rappelle celle de *salīva* et fait penser à un dérivé à redoublement **gen-g-īva*".[20] One can only agree with de Vaan's pronouncement (EDLIL s.v.) that the etymology of *gingīva* remains "unknown".

3 I suggest, however, that a more attractive hypothesis emerges if we consider the background of the Germanic words corresponding semantically (at least in part) to Lat. *gingīva*, such as Eng. *gum*(*s*) itself, as well as related material in Baltic.[21] There is widespread agreement that these words derive from the PIE root **ĝʰeh₂-* 'gape, be wide open' (a root recently treated in detail by Lubotsky, 2011: 107ff.), with the Germanic material based on its *u*-extended version **ĝʰeh₂-u-*.[22] Various unresolved morphological issues surrounding the Germanic and Baltic forms (including questions about ablaut and vocalism, and about the nature of their *m*-suffix or *m*-enlargement) can be left aside here.[23] The essential point is that PIE **ĝʰeh₂-*(*u-*)/**ĝʰh₂-*(*eu-*) 'gape' provides the basis for a well-established etymology for words for 'gum(s)' and 'palate' (in part also 'throat') in these languages, immediately attractive both formally and semantically.

16 See especially Schwyzer 1930.

17 Better than anything cited by Schwyzer is Skt. *danta-veṣṭa-* 'gums' (MBh.+; cf. *veṣṭate* 'clings to, dresses, envelops'), kindly brought to my attention by Werner Knobl.

18 We return below, however, to the possibility of an assimilation like **cingīva* > *gingīva,* which is not objectionable in and of itself.

19 These are: (i) a reduplicated formation based on PIE **ĝyeuH-* 'chew' (Osthoff and others); (ii) original **gimb-īva,* cf. *γαμφηλαί* 'jaws (of animals)' (Bersu); (iii) a formation in **gen-g-* with broken reduplication related to Lat. *gena* 'cheek' (Niedermann); and (iv) a form related to Westphalian *kinkel*(*n*) f. 'strip of flesh, double chin' (Sütterlin). See WH for critical remarks and bibliographical details.

20 For analysis in terms of "broken reduplication", see also André (1978: 47), though with a sympathetic discussion of Solmsen's solution.

21 OHG *goumo, guomo*, *giumo, gommo* 'roof of the mouth, palate', in part also 'throat' (NHG *Gaumen*), ON *gómr* 'palate; gums', Far. *gómi* 'oral cavity', OE *gōma* 'palate', pl. 'jaws' (Eng. *gum*(*s*)), all masc.; cf. (with velar for expected palatal, see next note for the traditional root shape) Li. *gomurỹs* m., *gomurẽ*/*gomerẽ* f. 'throat, gullet, Adam's apple', Latv. *gāmurs* 'larynx, windpipe'.

22 See, e.g., IEW 449 (s.v. **ĝʰēu-* 'gähnen, Klaffen'), Krahe-Meid 1969: 128, Winter 1982: 183, Bjorvand-Lindeman 2000: 311f. (s.v. *gomme*), Kluge-Seebold 2002: 334 (s.v. *Gaumen*), Lloyd-Lühr 2009: 563 (s.v. *goumo*), Watkins 2011: 32 (s.v. **ghēu**-), Kroonen 2011: 302ff., Kroonen (2013) s.v. **gōman-*/**gauman-*. Fraenkel (LEW 161, s.v. *gomurỹs*) hesitates, on account of the velar/palatal mismatch (see preceding note). To the same root probably also Lat. *faucēs* 'throat, larynx, pharynx', despite the etymological dictionaries; see Winter loc. cit.

23 I address these matters elsewhere, in a forthcoming paper that treats Lat. *famēs* 'hunger'.

4 According to Lubotsky's analysis (2011: 107), another enlarged form of this root is **ĝʰh₂-(e)n-*, as in Gk. *χάσκω* 'gape, yawn' (aor. *χανεῖν*, perf. *κέχηνα*) and Pr.-Gmc. **gana-* (ON *gan* n. 'yawning', cf. *gana* 'be opened [of eyes, mouth], stare').[24] One could thus, without much ado, set up a preform **ĝʰh₂(e)n-ĝʰ-* (with broken reduplication, cf. Meillet's comment cited in **2** above, with n. 20). The full grade **ĝʰh₂en-ĝʰ-* would presumably produce a form with *a*-vocalism, and thus itself is not usable; but the zero-grade option **ĝʰh₂n̥-ĝʰ-* is worth considering. An immediate difficulty is that such a preform, with its initial **ĝʰ-*, would yield Lat. *×hing(īva)*. Yet the distance from there (or from the preform **ĝʰh₂n̥-ĝʰ-*) to *ging(īva)* is not large. As shown by Kümmel (2012/2013), post-nasal voiced aspirates had already become voiced stops by the Proto-Italic period. Thus a preform **ĝʰh₂n̥-ĝʰ-* would yield, in the first instance, Italic **χeng-*. It would be simple to assume an early assimilation of this **χeng-* to **geng-* (or, at a later stage, pre-Lat. **heng-* > **geng-* or **hing-* > *ging-*). In a slightly more complex way, one could assume an assimilatory despirantization of **χeng-* to **keng-*, whence a further voicing assimilation of **keng-* to **geng-* (or, at a later stage, pre-Lat. **king-* > **ging-*); and indeed, as already discussed by Schwyzer (1930: 261n2), there is some support for "Fernassimilation" patterns of this kind in Latin, as with Gk. *Ἀκράγας* > *Agrigentum* and gloss forms of the type (*cingla* >) *gingla,* (*clangit* >) *glangit,* etc. (With a different syllabic structure, cf. also *nec legō* > *neglegō* 'disregard'; Leumann 1977: 199) Although distant assimilation does not require such a justification, note that the assimilation would have had the effect either of maintaining or producing a sense of the formation as reduplicated;[25] and there is always the possibility of analogical influence from related vocabulary, for which one suggestion will be offered below (**5** ad fin. with n. 32). One can, then, account in a relatively straightforward way for Lat. *ging(īva)* on the basis of an earlier **hing-* (e.g., from original **ĝʰh₂n̥-ĝʰ-*).

5 The most serious drawback of the analysis in **4** is that the enlarged root form **ĝʰh₂-(e)n-* is not actually attested in Italic, apart from the PIE word for 'goose', which Lubotsky reconstructs as **ĝʰh₂-ens-* 'gaper' (> Lat. *ānser*), based (according to this view) on the same root. What *is* attested, however, is the well-established "*i*-present" form of the root, i.e., **ĝʰeh₂-i-* (parallel to **ĝʰeh₂-u-*, as in the Germanic forms in **3**), with zero grade **ĝʰh₂-i-* and metathesized zero grade **ĝʰih₂-*: one or the other of these zero grades is the source of Lat. *hiāre* 'gape, be wide open' (Pl.+) and related material, such as *hīscere* and *hietāre* 'gape' (Pl.+), *hiulcus* 'open-mouthed' (Pl.+), etc. Elsewhere, *i*-present forms are found in Balto-Slavic (Li. *žióti* 'to open [one's mouth]', OCS *zijati* 'id.'), Tocharian (TB *kāy-* 'id.', pret. ptcple. *kakāyau*), and perhaps Indic (RV *jéhamāna-* 'panting, gasping'; Lubotsky 2012: 180f.).[26] Moreover, as Lubotsky points out (2011: 108), a nasal present built to the *i*-form of this root is securely reconstructable, based on evidence from Anatolian (Hitt. *kīnu-^{zi}/kinu-^{zi}/ginu-^{zi}* 'open [up], break open'), Slavic (Pr.-Sl. **zinǫti*: OCS *zinǫti,* SCr. *zȉnuti* 'open [one's mouth]', etc.), and Germanic (e.g., ON *gina*

24 Differently LIV², with a separate root **ĝʰan-* or **gʰan-* for this material; but see Lubotsky (2011: 107f.) for discussion, including arguments against LIV²'s conception **ĝʰeh₁i̯-* for the *i*-present form of the root (more on which below). LIV² Add., however, has now done away with **ĝʰan-* and **ĝʰeh₁i̯-* in favor of a basis in **ĝʰeh₂(i̯)-*, following Lubotsky.

25 Typologically similar (as pointed out to me by Adam Hyllested) would be Ved. *jihvā́-* 'tongue', if this results from an assimilated IIr. **dijʰwaH-* (← **dajʰwaH-*); see EWAia I.592 (s.v.).

26 Lubotsky operates with *i*-present formations characterized by Leiden-style suffix ablaut **-ei-/*-i-*, as opposed to the pattern with root ablaut (also admitting secondary full grades based on the zero grades) advocated by Jasanoff (2003: 91ff.; cf. Lubotsky 2011: 107 n6 and, contra suffix ablaut, Jasanoff 2009: 48f.). Since the issues surrounding Lat. *gingīva* mainly involve zero grades, this theoretical controversy is not directly relevant and can be safely ignored.

'gape, yawn', OE *gīnan* 'id.').[27] If one assumes, again, a form with broken reduplication, Lat. *gingīva* could be based on this nasal present stem, whether with zero grade **ĝʰih₂-i-n-ĝʰ-* or metathesized zero grade **ĝʰih₂-n-ĝʰ-* (the latter yielding Ital. **χīng-*, susceptible to Osthoff-shortening). The development **χing-* → *ging-* (instead of expected **hing-*) would then be handled as discussed in **4** above. The main deficiency of this version is the lack of any explicit account for the suffixation in *-īva*.[28]

One could also imagine an extended (perhaps "intensive") reduplicating pattern **ĝʰih₂-n-ĝʰih₂-* (> Ital. **χingī-*, after Osthoff-shortening): in that event, there would be no need to account for the full suffix formation in **-īu̯ā* – all that would be needed is suffixation in (Transponat) **-u̯éh₂* (Ital. **-u̯ā*), for which the simplest approach might involve modeling after 'tongue' (PIE **dn̥ĝʰ-uh₂*, but Ital. **dn̥g(u)u̯ā* > OLat. *dingua*).[29]

It is disquieting, however, that the nasal present stem, although plausibly reconstructable for PIE, is not attested in Latin. An alternative (and ultimately attractive) possibility, then, would be to consider the *-n-* of *gingīva* to be an adventitious insertion, which could thus have entered the word secondarily at any stage. The phenomenon of so-called "expressive nasalization" (including the subtype at issue here, with insertion of a nasal consonant) is well-known and well-documented.[30] Insofar as a reduplicated pattern may be involved here, a close parallel appears in Sanskrit intensives with nasal reduplication for roots lacking a nasal, e.g., *dandahīti* (BhāgP+) to *dah-* 'burn'. As discussed by Schaefer (1994: 54f.), the pattern seems to have begun in Vedic times, in part on the basis of false analogies: e.g., *jañjapyámāna-* (ŚB 11.5.5.10) to *jap-* 'whisper' (without etymological nasal), cf. *jañjabhyate* (TS, AV, etc.) to *jambh-* 'crush', in addition to the synchronic ambiguity of zero grades with *a*-vocalism, only some of which go back to forms with etymological nasal.[31] Similarly, in her detailed treatment of the phenomenon in Germanic, R. Lühr has shown (1988: 92–188) that many such instances (among them a number of Schallwörter, a prominent category of such forms) have arisen on the basis of analogies. Thus, without insisting on this particular scenario, it is possible that an original **ĝʰih₂-ĝʰih₂-* (with semantic reference to opening the mouth wide) came under the influence of the ancestor of Lat. *gingrīre* 'cackle (of geese)' (P.F. 84.12L), or even of **ĝʰans-* 'goose' itself (or, following Lubotsky, **ĝʰh₂-ens-*, perhaps even related to **ĝʰeh₂-i-*, as noted above).[32] As a further comparandum of some interest (kindly brought to my attention by Guus Kroonen), note OHG *gingēn* 'desire', a stative nasal-infixed pendant to Go. *ga-geigan* 'gain' — both generally assumed to derive from a *gʰ*-extended or reduplicated version of the (*i*-present form of the) 'gape' root (see, e.g., Lehmann 1986: 137, s.v. **ga-geigan*).

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27 Lubotsky (2011: 108n10) follows Kloekhorst's analysis of the Hittite spellings (2010: 215f.) and of the word's background as a *nu*-present **ĝʰh₂i-neu-*. Kloekhorst does not, however, address the plene spelling in the initial syllable (see his list of attestations, 2010: 216n54); as pointed out to me by Kazuhiko Yoshida, this feature may be old (note, e.g., *ki-i-nu-ud-du* KUB 30.10 obv. 24, 28; OH/MS), and thus remains unexplained under a reconstruction of the type suggested by Kloekhorst.

28 Further discussion of this point is reserved for **7** below.

29 Appeal to analogy with the suffix of *salīva* is hardly satisfying, since this, as already noted, is itself obscure.

30 For Latin, the treatment by Graur (1929: 94ff.) remains useful; note the extensive evidence for this pattern in late Latin and Romance (thus in the spoken register): Lat. *gibbus* 'hunchbacked' → Rom. **gimbus* (also **gubbus* and **gumbus*), Lat. *locusta/lōcusta/lucusta* 'locust; lobster' → Rom. **langusta,* Lat. *sabūcus/sābūcus/sabbūcus/sabuncus* 'elder(berry)' → Rom. *sambūcus,* etc.

31 Hence also (as pointed out to me by Werner Knobl) *agaṅgūyat* (PB 14.3.18) to *gu-* 'sound', and the adjective *apracaṅkaśa-* 'not looking forward' (AV-Ś 8.6.16a), based on the intensive stem of *kaś-* 'see' (cf. *cākaśīti,* RV+).

32 For *gingrīre*: there is even the possibility that such an association contributed to the assimilation of pre-Lat. **hing-* to *ging-* (or Ital. **χeng-* to **geng-*, etc.; cf. **4** above).

6 The proposals for the background of Lat. *gingīva* sketched in **4** and (especially) **5** incorporate a plausibility factor that is absent from previous approaches to the problem, given (a) the existence of words for 'gum(s)' in other Western IE languages based on variants of the same root (**3**), and (b) the fact that the proposed *i*-present basis **ĝʰeh₂-i-*, specifically in its zero grade form(s), is widely attested in Western IE languages and securely documented within Latin itself (**5**). But there is a modified version of this theory that may be more attractive and more interesting than the versions in **4** and **5**.

This begins from Lubotsky's reconstruction (2011: 108) of a *u*-stem noun **ĝʰh₂oi-u-* 'mouth' (in his notation), formed to the *i*-present **ĝʰh₂-ei-/*ĝʰh₂-i-* (in his conception, see n. 26): cf. Pr.-Toch. **koyə* (> TB *koyn* n., pl. *koynuwa* 'mouth', TA *koy-* 'id.') and Pr.-Sl. **zěvŭ* 'muzzle' (CSlav. *zěvŭ,* SCr. *zȉjev*).[33] Let us suppose, then, that the second part of the preform of Lat. *gingīva* – that is, the portion following one or another version of the nasal present (or secondarily nasalized) stem (as in **5**, i.e. **ĝʰh₂-i-n-* or **ĝʰih₂-n-*) – was an *eh₂*-derivative (abstract or collective) of this *u*-stem word for 'mouth', i.e. (**ĝʰh₂oi̯-u-* →) **-ĝʰh₂ei-u̯-éh₂*.[34] Apart from the wrinkles involving the (assimilated) initial consonant and Osthoff-shortening (**4**), such a form (i.e., **ĝʰh₂-i-n-ĝʰh₂ei-u̯-éh₂* or **ĝʰih₂-n-ĝʰh₂ei-u̯-éh₂*) would yield Lat. *gingīva* directly, given the regular Latin treatment of medial **-ei-* > *-ī-*.[35]

This account is attractive and potentially interesting for several reasons:

i Most obviously, it provides a full morphological accounting of the word, without recourse to "broken reduplication" and without having to posit a suffixation in "**-īu̯ā*" that is difficult to motivate except analogically.[36]

ii The word turns out to be a compound, for which, nevertheless, more than one analysis seems possible, with the precise details probably unrecoverable. Working backward, as it were, from the above hypothesis: if the word itself is an abstract or collective based on a verbal first member ('gape, open [the mouth] wide') and a nominal second member ('mouth'), the result may be interpreted as a concretization with a meaning like 'the opening-the-mouth[-wide] stuff', i.e., 'gum(s)' (or in principle also 'palate; jaw; throat'), these being the most salient items with such a descriptor, besides 'teeth' and 'tongue'.[37] This formation, in turn, could in principle have various derivational sources. Under the unmarked assumption that they, too, would originally have been compounds, one could imagine, for example, a bahuvrīhi source ('with open[-ed] mouth' → 'open-mouthed [stuff]'), or various types of verbal governing compound (e.g., 'opening the mouth' → 'open-mouthed [stuff]'). More fine-grained analysis is hampered by indeterminacies surrounding the first member: recall (**5**, ad fin.) that the *-n-* may either be part of the nasal present stem, or else adventitious; and in either case, there is at most a stem-form or what amounts to a

33 For TB *koyn*, TA *koy,* similarly Adams (1999: 203, s.v. *koyn*), who sets up a PIE **ĝʰoh₁i̯-u-* (neut.) to **ĝʰeh₁(i)-* 'yawn, gape' (cf. n. 24 above on the root shape).

34 The basic noun is best conceived of morphologically as an acrostatic *u*-stem (given its *o*-grade), for which a secondary derivative would naturally be based on its *e*-grade oblique stem. Cf. **dór-u* 'wood' → **der-u̯-éh₂*: Li. *dervà* (acc. *deřvą*) 'kindling'.

35 The absence of manuscript or inscriptional support for spellings like "*gingeiva*" is not problematic, given the relatively late and limited attestation of the word (**1**).

36 Again: more on the latter point in **7** below.

37 These latter, of course, were expressed with individual lexical items (well-established for PIE and inherited in Latin), without a need for compound or phrasal expressions. Given the Germanic material in **3**, it would in fact be reasonable to assume that *gingīva* originally meant 'palate, roof of the mouth', and (much as with OE *gōma* 'palate' > Eng. *gum*) was semantically adjusted to 'gum(s)' under competition from Lat. *palātum* 'roof of the mouth; vault' (Varro+), the background of which remains obscure (see EDLIL s.v.).

bare root (i.e., one of the zero grade allomorphs **ĝʰh₂i-* or *ĝʰih₂-*).[38] This, however, raises another possibility, namely that the sequence in question reflects not an original compound, but rather a univerbation. As is well known, univerbations often show the effects of allegro phonology, leading to first-member reductions, as in Lat. *īlicet, scīlicet,* etc. (for original *īre licet, scīre licet,* etc.).[39] We could thus imagine, for example, a univerbated imperative phrase 'open the mouth!', with truncated imperative ending (whether **-dʰi* or **-e,* depending on the underlying verbal formation),[40] whence compounded 'open-mouthed' and 'open-mouthed [stuff]'.

iii What is particularly noteworthy about this sequence is that both components are based on the same root: in other words, the expression, whether conceived of as an original compound or univerbation, is a *figura etymologica.* In itself, this is not particularly striking, even if relatively unusual as a pattern of word formation. One thinks of instances like Eng. *sing-song*,[41] and Werner Knobl has kindly drawn my attention to Gk. *θεσμοθέτης* 'lawgiver' (Aristoph.+), as well as a number of similar formations in Vedic language.[42] What renders this analysis of *gingīva* potentially remarkable is the fact that, as observed by Lubotsky (2011: 108),[43] the TB collocation *koyn kāy-* (see above on *koyn* and **5** on *kāy-*) is a *figura etymologica,* and is attested as a fixed expression meaning 'open one's mouth (wide)'.[44] If virtually the same expression underlies Lat. *gingīva* etymologically (where the major difference, apart from the nominalization itself, consists in the use of an *i*-present-based nasal stem, rather than the plain *i*-present stem, as in TB *kāy-*), we might have evidence for a Latin/Tocharian "phrasal isogloss", perhaps reflecting a shared archaism of considerable antiquity. The fact that the verbal element in *gingīva* appears in first position (according to the above analysis) does not seem problematic: this could be a normal feature of an original imperative phrase (cf. RV *píbā sómam* 1.15.5b, etc., beside *sómaṃ píba* 1.15.1a, etc.).

iv A final point concerns the original meaning of the verbal root itself. As T. Pronk has observed (apud Kloekhorst 2010: 217n55), the evidence of Hitt. *kīnu-ᶻⁱ* 'open [up]' and of TB *kāy-* 'open wide' suggests that "the original meaning of **ĝʰh₂(-ei)-* was 'to open up', and that the semantic development to 'to open one's mouth, to yawn' is a common innovation of the other IE languages". This may be a slight exaggeration, as a trace of the broader meaning may also appear in Gk. *χά(ϝ)ος* 'primordial Chaos (Hes. *Th.* 116); empty space, void (Aristoph.+)', cf. *χαῦνος* 'empty, porous'.[45] Still, if Lat. *gingīva* continues the same *figura etymologica* as in TB *koyn kāy-,* it would thereby preserve another trace of the archaic meaning of the verbal root – more archaic, semantically, than the oral specialization seen in Lat. *hīscere, hiāre,* etc.

38 Cf., for example, the root-like behavior of the latter in Lat. *hīscere* < *ĝʰih₂-sk̂é/ó-*.

39 Leumann 1977: 566. For recent discussions of univerbation and consequent allegro reduction in the context of IE phraseology, see Hackstein 2010 (passim), 2012a: 88f., 2012b.

40 Or possibly even endingless, depending on the antiquity of the endingless type, as in OLat. *ei* 'go!', FU (Carm. Arv.) 'be(come)!', etc.

41 Attested first as noun (early 17th c.+), then adjective (late 17th c.+), and finally verb (18th c.+).

42 To consider only RV material: *dhāma-dhā́-* 'lawgiver', *bhāra-bhṛ́t-* 'bearer of a burden', *soma-sút-* 'soma-pressing', *sutá-soma-* 'with pressed soma'.

43 And (as he points out) as already noted by Adams (1999: 203, s.v. *koyn*).

44 For typical contexts, see the passages cited at Adams 1999: 153 (s.v. *kāy-*). On TB *koyn kāy-* (and for *figurae etymologicae* in Tocharian generally), see Malzahn 2010: 98.

45 See GEW s.v. *χάος* for semantic analysis and further references.

7 The suffixal formation of Lat. *salīva* emerges in a new light, now that (according to the theory offered in **6**) the ending of Lat. *gingīva* would not reflect influence from *salīva,* but has its own organic morphological explanation. Accounting for the suffixation of *salīva* itself is in fact quite difficult, even though the older literature has nothing at all to say on the subject.[46] More recently, de Vaan (EDLIL 536, s.v.) observes that if the word belongs with the **solH-uā̆-* '(yellowish) grey' of Balto-Slavic and Germanic,[47] one could imagine an original (trisyllabic) **saluus,*[48] for which (says de Vaan) one might compare *irrigīvus* 'well-watered' beside *irriguus* 'irrigated' or *vocīvus* beside *vacuus* 'unoccupied'. But – even apart from the fact that there is no trace in Latin of the presumed *u*-adjective **solH-u-,* much less a thematized version of it – Lat. *-uus* and *-īvus* forms like these are not comparable: both involve deverbal *-uus* (of the type *caeduus* 'for cutting' ← *caedō* 'cut', etc.) and deverbal *-īvus* (*irrigīvus* ← *irrigō, vocīvus* ← *vacō*), neither of which is relevant here; and for denominal *-īvus*, the productive type in *-t-īvus* based on *-to-* adjectives (*captus* 'taken' → *captīvus* 'captive', etc.) is likewise clearly of no direct use. That leaves the extremely rare synchronically denominal-looking type *internecīvus* 'fought to the death' (Cic.) ← *interneciō* 'total destruction' (Lucil.); but this is probably also deverbal in origin, insofar as the basis is a verbal abstract, and such forms generally involve nominal formations with suffixal *-i-* (such as *-iō* or *-iēs* nouns) in any case.[49] This again offers no clear support for *salīva.* The first thing to conclude is that Lat. **-īu̯o-* is not clearly derivable within Latin, and thus one must seek guidance from comparative material.[50] This is found principally in the Slavic adjectives in *-ivŭ,* which have two main sources: a deverbative type is transparently based on *i*-verbs (OCS *ljubiti* 'love' → *ljubivŭ* 'loving', etc.), and a denominative type is transparently based on *i*-stem nouns (OCS *rěčĭ* 'speech' → *rěčivŭ* 'talkative', etc.); the latter thus invites analysis as a "deinstrumental" formation in **-i-h_1* + (adjectival suffix) **-u̯o-*.[51] The second thing to conclude is that the **solH-u-* material confers no actual advantage towards explaining the suffix formation of *salīva*: *i*-stems are entirely lacking – apart from the **solH-u-* material itself, one finds only root formations or other suffixed formations.[52] Under the circumstances, then, there is not much reason to prefer the traditional etymology of *salīva* based on **solH-u-,* if a better alternative can be found.

46 Meillet (EM 590, s.v.) merely observes "[m]ême formation que *gingīva*" and renounces any attempt at etymology; WH (II.468, s.v.) supports a connection with material in Celtic, Germanic, and Balto-Slavic meaning 'dirty; dirty-colored' (more on which shortly), but says nothing about the suffix formation; Leumann (1977: 304) lists *salīva* together with *gingīva* in a "relic" category of *-īvus* forms, without further comment.

47 Essentially the WH etymology (cf. preceding note); the material in question includes Russ. *solóvyj* 'light bay', *solovój* 'yellowish gray', OHG *salo* (gen. *salwes*) 'dirt', OIc. *sǫlr* 'dirty yellow'.

48 Presumably (although de Vaan does not spell this out) based on a thematized version of **solH-u̯-*, either **sḷH-eu̯-o-* or root-accented **sḷ́H-u̯-o-* (> **salau̯o-*). The latter, to be sure, would run counter to Leiden School doctrine (see, e.g., Schrijver 1991: 193ff. and EDLIL passim for relevant items [*calamitās, palma,* etc.], vs. Nussbaum 1997: 186n42, 197 [Lat. *palma*], Meiser 1998: 108f., Weiss 2011: 110); but this is of no consequence here.

49 On all these formations, see Leumann 1977: 303ff., Weiss 2011: 298f.

50 Similarly Nussbaum 1975: 131 on Lat. *-īvus* ("entirely defunct at an early stage") and on the generally late character of Lat. (*-to-* →) *-tīvus.* (Differently on *-tīvus,* and without consideration of plain *-īvus*: Rasmussen 2005.)

51 On deinstrumental formations in Latin nominal derivation, see Weiss 2011: 290 (*-ūno-, -ōno-*), 293 (*-ītus, -ūtus, -ōtus*), 296 (*-īko-*).

52 Thus OIr. *sal* 'dirt' (f. *ā*) and similar Celtic material (if it belongs here at all, cf. EDP-C s.v. **salā*); Lat. *salebra* (Cic.+, mostly plural) 'roughness, harshness, unevenness' (< **sal-es-ro-*) is placed here by de Vaan, but may rather belong with *saliō* 'jump' (thus EM 590, s.v. *saliō*).

8 These circumstances warrant a reconsideration of the older etymology of *salīva,*[53] comparing instead words for 'slime' and the like, including Lat. *līmus* 'mud, slime', OIc./OE/MHG *slīm* 'slime', and particularly OCS *sliny* (pl.) 'saliva' (cf. Russ. dial. *slína/sliná*, SCr. *sl̓ina,* Latv. pl. *sliēnas* 'id.'). Note that Lat. *salīva* and its derivatives can in fact refer to viscous or unpleasant liquids generally, and not just 'spittle', a usage that may well be older than its first appearance in Augustan literature (Prop. 4.7.37 *salīvās* 'poisons', cf. *salīvōsus, salīvārius* 'slimy, clammy' [Pliny+]).

The formal details must remain to some degree unrecoverable, just as with the similarly isolated term *gingīva*; and there is space here only for the following tentative sketch. The Balto-Slavic intonation (cf. also Li. *slíekas* 'earthworm') points to a laryngeal root of the type **sleiH-* or **sleH-i-*;[54] and a zero grade **sl̥H-i-* (to **sleH-i-*) would provide an *i*-stem basis resulting in (pre-Lat.) **sali-*. Assuming substantival meaning, the instrumental singular of such a form (**sl̥H-i-h₁* 'with slime' > pre-Lat. **salī*) could have been renewed as an adjective **salī-u̯o-* 'slimy',[55] with a feminine substantivization **salī-u̯ā* 'slimy stuff, saliva' that is either a "real" formation,[56] or else one that has undergone influence from the apparent (though not real, according to the above theory) labial suffixation in *gingīva* (at the stage **gingeiu̯ā* or still earlier) and/or the other prominent oral-cavity term, pre-Lat. **dinguu̯ā* 'tongue' (cf. **5** above), an analogical process that could have occurred at any stage beginning with a Proto-Italic **salīu̯ā* (beside **χingeiu̯ā, *dn̥guu̯ā*).[57]

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53 See Vaniček 1881: 341f. (with earlier references); similarly (though incoherently) Muller (1926: 399), who operates with several roots at once.

54 See EDSIL 453 for the Balto-Slavic 'saliva' material, for which Derksen sets up **sleh₁i-n-eh₂*; similarly EDLIL 342 (s.v. *līmus* 1) for the root shape.

55 I.e., in the usual "deinstrumental" way (cf. n. 51 above), and essentially in the manner of Slav. *-ivŭ*.

56 I.e., roughly akin to a form like *scobīna* 'scraper, rasp' (: *scobis* 'sawdust, shavings'), among other *-īna* substantivizations to *i*-stems (Leumann 1977: 327f.), some of which are indeed elliptical (cf. Solmsen's alleged *salīva* [scil. *aqua*], n. 9 above). Note that I do not assume regular metathesis in a (quasi-)PIE form like **sl̥H-i-h₁*; as pointed out to me, however, by Jens Rasmussen, those who do could account for Lat. *salī-* by positing a "Lindeman variant" **sl̥lī* beside the metathesized **slī*.

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